

Artist Controlled Fracture Design Using Impurity Maps

Avirup Mandal
avirupmandal@ee.iitb.ac.in
IIT Bombay
Mumbai, Maharashtra, India

Parag Chaudhuri
paragc@cse.iitb.ac.in
IIT Bombay
Mumbai, Maharashtra, India

Subhasis Chaudhuri
sc@ee.iitb.ac.in
IIT Bombay
Mumbai, Maharashtra, India



Figure 1: An interactively created impurity map adds shallow lines of impurities on a pizza mesh as shown in the left image. When pulled, physically accurate simulation ensures that the cheese breaks in stretchy bits, while the pizza slice comes apart along these lines, as desired by the artist.

ABSTRACT

When an object breaks, simulating evolution of fracture as per artist control while maintaining physical realism and plausibility is a challenging problem due to different complex material properties of real world objects. In this work, we present impurity maps as a way to guide fracture paths for both brittle and ductile fracture. We develop a novel probabilistic damage mechanics to model fracture in materials with impurities, using a random graph-based formulation in conjunction with graph-based FEM. An artist created map allows us to selectively distribute the impurities in the material, to weaken the object in those specific regions where the imperfections are added. During simulation, the presence of impurities guide the cracks that develop such that the fracture pattern closely follows the impurity map. We simulate artist-controlled fractures on different materials to demonstrate the potency of our method.

KEYWORDS

fracture, random graph, impurity, artist

ACM Reference Format:

Avirup Mandal, Parag Chaudhuri, and Subhasis Chaudhuri. 2022. Artist Controlled Fracture Design Using Impurity Maps. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Posters (SIGGRAPH '22 Posters)*, August 07-11, 2022. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3532719.3543202>

1 MOTIVATION

Artist controlled design of fracture that still behaves in a physically consistent manner is a difficult problem. Since state or condition of material affects the fracture generation and propagation, we present a method to selectively weaken a material by corrupting

it with impurities along artist directed paths or regions. However, objects made of such non-homogeneous materials display uncertain and difficult to model fracture characteristics. Purely deterministic approaches [O'Brien and Hodgins 1999] [Wolper et al. 2020] with pre-defined fracture thresholds cannot model such phenomena. We propose a novel probabilistic damage mechanics model to model changes in material properties due to the presence of impurities. Our method lets the artist interactively create an *impurity map* on the object, much like a texture map. A surface map can be propagated inwards into the object, akin to a volumetric texture. Impurities get distributed into the material as per the impurity map and weaken the material accordingly. When the object breaks, the fracture pattern closely follows the distribution of impurities as encoded by the impurity map. We use a random graph-based probabilistic approach to simulate the damage of impure materials. We extended the prior works on graph-based FEM [Khodabakhshi et al. 2016] [Mandal et al. 2021] to develop our method.

2 PROBABILISTIC DAMAGE MECHANICS

In graph-based FEM, cracks due to fracture are generated depending on the stress value along the direction of an edge contained in an underlying graph structure. This graph structure forms the base on which we apply a random graph [Pittel 2010] model to incorporate probabilistic threshold criteria.

2.1 Random Graph

Let us define a random graph growth process $\{G(n_v, \mu)\}$ on a vertex set $[n_v]$ with μ successive steps. Let $G(n_v, 0)$ be an empty graph. Recursively, given a graph $G(n_v, \mu)$ of vertex degree $\mathbf{d} = \{d_1, d_2, \dots, d_n\}$, a new graph $G(n_v, \mu + 1)$ is obtained by inserting a set of new edges between any two still disjoint random nodes i and j with $i \neq j$. The probability that an edge e will join two currently disjoint vertices, $i \neq j$, is proportional to $(d_i + \alpha)(d_j + \alpha)$. Here d_i, d_j are the degrees of i, j in $G(n_v, \mu)$ and $\alpha > 0$ is a user defined constant. Degree of any node refers to the number of other nodes it is connected to. The sequence $\{G(n_v, \mu)\}$ is a Markov process.

A multigraph process is defined as a sequence $\{MG(n_v, \mu)\}$ of multigraphs where self-loops and multiple edges on a single vertex are allowed. Given a current multigraph, $MG(n_v, \mu)$, with degree $\mathbf{d}$

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SIGGRAPH '22 Posters, August 07-11, 2022, Vancouver, BC, Canada

© 2022 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-9361-4/22/08.

<https://doi.org/10.1145/3532719.3543202>

on the vertex set n_v , a new edge joins two vertices, i and j , with probability proportional to $2(d_i + \alpha)(d_j + \alpha)$ and forms a loop $i \rightarrow i$ with probability proportional to $(d_i + \alpha)(d_j + \alpha + 1)$. Note that in multigraph while $i \neq j$, they may necessarily not be disjoint. This random process forms a Markov chain whose state space is the set of all multigraphs on the vertex set n_v .

Putting it in our context, we start from an intermediate multigraph, $MG(n_v, \mu)$, with degree $\mathbf{d} = \{d_1, d_2, \dots, d_n\}$, on a vertex set of n_v nodes. Let us assume that the structure of the intermediate multigraph which contains n_v nodes and n_e edges, coincides with the initial undamaged graph generated by our FEM simulation mesh model. Then graph-based FEM for fracture simulation can be conceived as removing edges from a graph, contrary to addition of edges as explained above. However, it still remains a Markov chain whose state space is the set of multigraphs. Given any multigraph, $MG(n_v, \mu)$ with degree $\mathbf{d}$, we define the probability to remove an edge between vertices i and j as

$$p_{ij} \propto \left\{ 1 - h \left[2 \left(d_i + \frac{1}{|\alpha_i|} \right) \left(d_j + \frac{1}{|\alpha_j|} \right) \right] \right\} \quad (1)$$

where α_i and α_j are random but constant initial values associated with node i and j . Function h maps the random numbers from sample space to probability space. Note that we have not included the probability of self-loop in Equation 1 as it does not occur in a consistent FEM mesh. The impurities are themselves modelled as additive white Gaussian noise, added to the nodes of the object mesh at the start of the simulation. High values of α_k (for any node k) denotes more potent impurities. Function h can be any strictly increasing function with range $[0, 1]$.

As evident from Equation 1, if the potency of impurities increases, i.e., if we have higher values of α_k , the function value goes down. This, in turn, makes fracture probability of the corresponding edge increase. Zero value of α_k corresponds to a sharp cut.

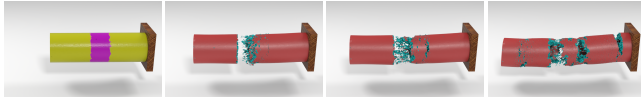


Figure 2: Illustration of impurity map (left). Effect of the impurity is decreased from left to right (next three).

2.2 Impurity Induced Fracture

In Figure 2, we show the effect of impurities on fracture. We take a hollow cylindrical tube and add an impurity map to it. The impurity map is displayed on the left most image of Figure 2. The impurity map shows the location of the impurities (in magenta) added to the object. The tube is then hinged at one end and pulled from other. In the next three images of Figure 2, the α_k value is decreased from left to right. Higher values of α_k cause more weakening of the material. As evident from Figure 2, decreasing value of α_k lessens the effect of impurities on the overall strength of the object and causes fractures in multiple locations in the tube as opposed to only in the region where impurities have been added.

3 ARTIST CONTROLLED FRACTURE DESIGN

We develop an interactive interface to allow artists to design any specific fracture patterns that they want, using impurity maps.

Using our interface, an artist can control a virtual sculpting tool to add the map to an object mesh. When the tool intersects an edge, we weaken the fracture threshold of that edge by adding impurity to the two nodes of the edge. The artist can also control the potency of the impurity being added. During simulation when the fracture propagates, it closely follows the impurity map.

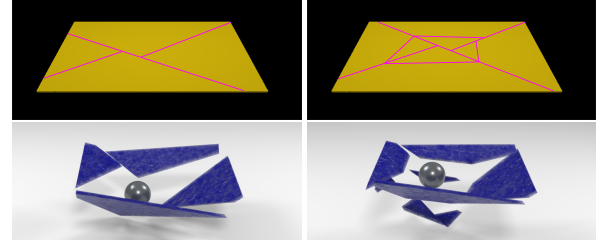


Figure 3: Impurity maps added to the glass slab model (top row). Upon impact, the glass slab fractures along regions defined in the impurity maps (bottom row).

In Figure 1, we show the impurity map added to a pizza model. As shown in the figure (leftmost), when the magenta regions in the impurity map touch or pass through any edge in the object mesh, the nodes corresponding to that edge gets a user-controlled α_k value assigned to them. Like a real life pizza maker who makes shallow cuts on the pizza to make slices but does not completely separate them, we weaken the pizza along these shallow deposits of impurities. When the pizza slice is pulled apart in Figure 1 (2nd on left to last), it closely follows the impurity distribution regions from the impurity map, but at the same time generates stretchy bits of cheese. This shows the ability of our framework to allow the design of ductile fractures.

In Figure 3, we embed an impurity map on a glass slab (magenta lines). When hit by a metal ball, the slab produces sharp shards or fragments that cleanly break along the fracture edges defined by the pattern. This clearly illustrates the ability of our method to produce artist controlled brittle fractures as well.

4 CONCLUSION

We introduce a novel probabilistic damage mechanics formulation to incorporate the uncertainties during fracture, caused by the presence of material impurities. We leverage this formulation to allow artists to interactively design impurity maps. These maps are then be used to guide the evolution of brittle and ductile fracture in the object, when it breaks during simulation.

REFERENCES

- P. Khodabakhshi, J. N. Reddy, and A. Srinivasa. 2016. GraFEA: a graph-based finite element approach for the study of damage and fracture in brittle materials. *Meccanica* 51 (2016), 3129 – 3147.
- A. Mandal, P. Chaudhuri, and S. Chaudhuri. 2021. Remeshing-Free Graph-Based Finite Element Method for Ductile and Brittle Fracture. *arXiv:cs.GR/2103.14870*
- J. F. O'Brien and J. K. Hodgins. 1999. Graphical Modeling and Animation of Brittle Fracture. In *Proc. of SIGGRAPH '99*. 137–146.
- Boris Pittel. 2010. On a random graph evolving by degrees. *Advances in Mathematics* 223, 2 (2010), 619–671.
- Joshuah Wolper, Yunuo Chen, Minchen Li, Yu Fang, Ziyin Qu, Jiecong Lu, Meggie Cheng, and Chenfanfu Jiang. 2020. AnisoMPM: Animating Anisotropic Damage Mechanics. *ACM Trans. Graph.* 39, 4 (2020), 16.